

The Swarm Simulator: A Dynamical Systems Model of Collective Intelligence Using the TO/TOGT Operator Pipeline

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Abstract

We introduce the Swarm Simulator, a multi-agent dynamical system based on the generative operator pipeline $G = U \circ F \circ K \circ C$ from Topographical Orthogonal Generative Theory (TO/TOGT). Each agent undergoes local compression, curvature intensification, loss of injectivity, and stabilisation. At the collective level, four operators—shared-intent stability I_t , coordination efficiency C_t , type-propagation multiplier M_t , and diffusion factor F_t —govern the swarm’s evolution.

All mathematical results in this paper follow directly from previously proved theorems in the Principia Orthogona series: fixed-point theory, contraction mapping, saturated pitchfork bifurcation, and the algebraic separation theorem (HAL hal-05555216v1). No empirical claims, numerical simulations, or unverifiable assertions are made. All formulas are presented in a “do not trust, verify” manner.

This work is mathematically independent of speculative physical or cosmological models that use similar terminology. Here, “orbit” refers strictly to operator-generated identity trajectories in dynamical systems.

MSC codes: 37C25, 37D10, 37G10, 47H10, 68T99.

1 Relation to Prior Work

The mathematical study of multi-agent dynamical systems has produced rigorous models of flocking, consensus, and emergent collective behavior. Foundational work by Cucker and Smale [6] and Olfati-Saber [5] established convergence guarantees for distributed agents interacting through local rules. More recent formulations include overdamped Langevin models for swarm cooperation [1] and minimal dynamical systems exhibiting simultaneous flocking and swarming [2].

Contraction theory has become a central tool for proving global convergence in multi-agent systems. Rivera-Ortiz et al. [3] provide explicit contraction-based guarantees for pursuit–evasion dynamics, while Leonard [4] surveys bifurcation structures—including pitchfork and Hopf bifurcations—in collective migration and decision-making. Pitchfork bifurcations in distributed decision networks have been rigorously derived in honeybee-inspired models [7].

The present work differs from these models in that the Swarm Simulator is not a physical or biological model but an operator-algebraic construction derived from the TO/TOGT generative pipeline. The contraction and fixed-point results used here follow directly from previously proved theorems in the Principia Orthogona series, and the four collective operators (I_t, C_t, M_t, F_t) provide an algebraic decomposition that unifies the stability, coordination, propagation, and diffusion dynamics found across the literature.

2 Swarm State Space

Definition 2.1 (Swarm State). *The swarm state at time t is the tuple*

$$X_t = (I_t, C_t, M_t, F_t),$$

where:

- I_t : shared-intent stability,
- C_t : coordination efficiency,
- M_t : type-propagation multiplier,
- F_t : diffusion factor.

3 Collective Operators

Definition 3.1 (Stabilising Operator).

$$I_{t+1} = I_t \cdot f_{\text{types}} \cdot f_{\text{agents}} \cdot (1 - \eta_t).$$

Definition 3.2 (Coordination Operator).

$$C_{t+1} = C_t^{\text{raw}} \cdot I_{t+1} \cdot \frac{1}{1 + D_t}.$$

Definition 3.3 (Type Propagation Operator).

$$M_{t+1} = M_t (1 + \beta \cdot \text{reuse}_t) \cdot \text{avg_quality}.$$

Definition 3.4 (Diffusion Operator).

$$F_{t+1} = 1 + \alpha t.$$

4 Composite Operator

Definition 4.1 (Swarm Evolution Operator).

$$G_{\text{swarm}} = F \circ M \circ C \circ I.$$

Proposition 4.1 (Dynamical System).

$$X_{t+1} = G_{\text{swarm}}(X_t).$$

5 Contraction and Fixed Points

Theorem 5.1 (Contraction). *If the coupling between agents satisfies*

$$L_I + L_C + L_M < 1,$$

then G_{swarm} is a contraction on $\mathbb{R}^4$ with the ℓ^1 norm.

Theorem 5.2 (Unique Fixed Point). *Under the contraction condition,*

$$X^* = G_{\text{swarm}}(X^*)$$

has a unique solution.

Theorem 5.3 (Global Convergence). *For any initial state X_0 ,*

$$\|X_t - X^*\| \leq L^t \|X_0 - X^*\|.$$

6 Multi-Orbit Emergence

The multi-orbit identity theory (Zenodo DOI: 10.5281/zenodo.19208015) implies that when multiple agent clusters satisfy the contraction condition independently, the swarm forms a multi-orbit system

$$\mathcal{S} = \{\mathcal{O}_1, \dots, \mathcal{O}_k\},$$

with higher-order invariant

$$\text{Inv}(\mathcal{S}) \supsetneq \bigcup_i \text{Inv}(\mathcal{O}_i).$$

7 Minimal Simulator

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1 def step(state, params, t):
2     I, C, M, F = state
3     p = params
4
5     I_new = I * p.type_quality * p.agent_quality * (1 - p.noise)
6     C_new = C * I_new / (1 + p.drag)
7     M_new = M * (1 + p.beta * p.reuse) * p.avg_quality
8     F_new = 1 + p.alpha * t
9
10    return (I_new, C_new, M_new, F_new)

```

8 Discussion

Classical models such as Cucker–Smale [6], Olfati-Saber [5], and Langevin-based swarm cooperation [1] demonstrate that global order can emerge from simple local rules. Similarly, bifurcation analyses in collective decision-making [4, 7] show how multi-stable states arise from symmetry-breaking interactions. The Swarm Simulator recovers these qualitative behaviors, but through a single algebraic operator pipeline derived from TO/TOGT. Rather than proposing a new physical model, this work provides a unifying operator-theoretic structure that yields contraction, fixed-point, and multi-orbit guarantees across all four collective quantities. All results follow from previously proved theorems; no empirical predictions are made.

References

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